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CASE REPORT

Climate threat and adaptation strategies in Cappadocia: A comparative analysis for Nevşehir

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ABSTRACT

Despite being one of the most special regions of Türkiye with its natural and cultural heritage, Nevşehir faces threats such as climate change, environmental pressures and unplanned urbanization. Therefore, a comprehensive and strategic roadmap is needed for sustainable urban and rural transformation. The renewable energy potential of the region offers a significant advantage for sustainable development. The establishment of solar power plants will encourage energy-efficient building designs and reduce carbon emissions by supporting local energy cooperatives. In addition to providing environmental benefits, these projects will also contribute to economic development by creating employment. Organic production should be encouraged in agriculture, modern irrigation systems should be expanded and water saving policies should be adopted. Economic incentives for cooperatives will support rural development by increasing the income level of farmers. In addition, ecological tourism practices and the management of cultural heritage areas within the framework of sustainable tourism principles will preserve the long-term attractiveness of the region. Prioritizing green infrastructure projects will improve the urban microclimate and provide a healthier environment. Shaping sustainability goals with public participation will strengthen social harmony. Nevşehir's adaptation to climate change while preserving its natural and cultural values will make the region an exemplary model in the national and international arena.

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INTRODUCTION

The effects of global climate change are caused by the increase in the global average temperature of the earth by approximately 1.1 ºC compared to the 1850 - 1900 years as a result of the increase in the accumulation of greenhouse gases in the atmosphere [1]. In particular, the increase in fossil fuel consumption, deforestation, the expansion of industrial production processes and intensive agricultural activities have caused a significant increase in greenhouse gas emissions [2]. This has triggered an increase in global average temperatures. Greenhouse gases such as carbon dioxide (CO_2), methane (CH_4) and nitrogen oxide (N_2O) accumulated in the atmosphere have had a direct impact on the Earth's energy balance, causing global warming to accelerate [3]. As a result, disruptions have occurred in the hydrological cycle, the frequency and severity of extreme weather events such as droughts and floods have increased and thus, biodiversity has faced serious threats [2]. The weakening of the resilience of ecosystems has put pressure on critical sectors such as agriculture, energy and water management, and issues such as food security and sustainable management of water resources have become priority issues on a global scale. Climate change is not limited to environmental impacts. It also causes radical transformations in the socio-economic structures of societies, reshaping rural and urban living conditions and threatening economic stability [4,5].

The regional effects of climate change vary depending on geographical location, topography and ecosystem dynamics. These effects become much more evident, especially in climate-sensitive regions. Türkiye is among the countries with high vulnerability to climate change due to its location in the Mediterranean basin. Scientific studies conducted in recent years have revealed that temperature increases are accelerating throughout Türkiye, serious irregularities are occurring in the precipitation regime and extreme weather events (e.g., long-term droughts, extreme heat waves, sudden downpours and floods) are increasingly frequent and intense [6, 7]. Especially in regions that are not rich in water resources, these changes lead to a decrease in groundwater levels, a decrease in agricultural productivity and a disruption in the hydrological balance [7]. The continental climate zone, which covers a large area in Türkiye, is among the regions most affected by these changes. The environmental and

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economic risks caused by climate change are also discussed in this study.

The province of Nevşehir and the Cappadocia Region examined within the scope of this study have the typical continental climate characteristics of Central Anatolia. However, the temperature increases, decreases in precipitation and the increase in the frequency of extreme weather events in recent years pose serious threats to the ecological and economic balance of the region [6]. While the increase in evaporation rates due to the increase in temperature in Nevşehir and its surroundings has negative effects on agricultural production, the decrease in water resources directly affects the rural development of the region. Traditional viticulture and agricultural production activities are at risk due to irregular weather conditions, water scarcity and sudden temperature fluctuations caused by climate change. However, in the Cappadocia Region, which has great economic value in terms of tourism, the protection of natural and cultural heritage assets is becoming increasingly difficult. Fairy chimneys and other geological formations are negatively affected by temperature changes and heavy rainfall [8], and ecotourism activities in the region are affected by changing visitor trends due to climate change. Increased extreme temperatures and prolonged drought periods may shorten tourism seasons and increase the fragility of the regional economy. Therefore, scientifically evaluating the effects of climate change on Nevşehir and Cappadocia Region and increasing the adaptation capacity of the region to these changes is of vital importance in terms of ensuring environmental sustainability.

The main purpose of this study is to evaluate the possible effects of climate change in the urban and rural areas of Nevşehir and to develop sustainable plans and strategies to combat these effects. For this purpose, the environmental, economic and urbanization problems of the city were also evaluated within the scope of the study. Since the Cappadocia Region contains religious, natural, historical and cultural assets that are also included in the UNESCO World Heritage List, this study also aims to create a roadmap by focusing specifically on the protection of heritage assets against climate threat. Therefore, concrete suggestions were also presented within the scope of the study within the scope of combating climate change in the Cappadocia region. In addition, in order to improve the welfare level in the city and continue sustainable development, initiating a climate-friendly transformation in the tourism sector, expanding environmentally friendly applications in urbanization and restructuring agricultural activities with sustainable modern methods are among the priorities of this study.

MATERIALS AND METHOD

In this study, Nevşehir province, which is located in the middle of Türkiye, was chosen as the study location due to both its vulnerability to climate change and the natural, religious, cultural and historical heritage assets in the city. The map showing the province of Nevşehir is presented in Figure 1.

The study was conducted in two phases. In the first phase, a large number of scientific studies and reports on climate change and its impacts, combating climate change, climate change adaptation strategies and the effects of climate change on heritage assets were examined. In the second stage, on-site inspections and investigations were carried out in 2024 to determine the general situation of the city of Nevşehir and the consistency of the proposed measures. Thus, the general problems and vulnerability of the city were identified and literature-supported measures were proposed.

Figure 1. The map showing the province of Nevşehir in Türkiye

Climate Change in Nevşehir, Türkiye

Drought is one of the most vital problems associated with climate change [20]. The threat of drought continues to increase each year as a result of reducing precipitation and increasing temperatures every year in the Cappadocia region. Water resources in Nevşehir are almost totally reliant on precipitation, making them vital to agricultural activities in the region [6]. Climate change, however, inhibits renewal of these resources. High evaporation and irregular precipitation have a direct effect on agricultural production, hence jeopardizing farmers' incomes and the economy of rural areas. Drought impacts the whole structure of soil. The reduction of moisture in soil causes accelerated erosion in the region, thus ending up losing fertile agricultural lands. Not only does this hinder the agricultural sector, it also threatens biodiversity and the ecosystem services in the region.

The water resource in Nevşehir poses the largest threat for agricultural production in the region. Withdrawn underground and decreasing surface water resources give place to decreased supply, thus making the climate problem of water scarcity much more visible [21]. Climate change affects these resources by further sampling them and revealing the scarcity. Water shortage mainly concerns the agricultural sector. Agriculture itself is getting insidiously fragile as the irrigation water demand continually increases with the declining resource base. Water scarcity affects the efficiency and diversity of agricultural production. Additionally, it places higher costs on the farmer's hands, being much difficult to small-scale farmers with higher irrigation costs and lower production efficiencies. This poses a threat to agricultural sustainability in the region. Decreasing water resources affect the animal husbandry activities practiced within the region. Decreased water for drinking and agricultural irrigation largely impacts rural populations' livelihoods.

The increasing number of extreme weather events is one of the most important evidences of climate change [22]. Such catastrophes include severe heat waves, sudden heavy rainfall, hail and storms, which severely damage the tourism sector as well as settlements in Cappadocia. For instance, these fierce heat waves would shorten the length of stay of tourists in the region, causing a decline in tourism revenues. Heavy rains and flooding accelerate the erosion of sensitive natural formations such as fairy chimneys and valleys, as well as damage them as heritage assets. Storms and hail threaten tourism infrastructure. Because they have overriding effects on historical buildings and outdoor activities. Particularly, such untimely events in rural areas also inflict economic damage. Decline in agricultural production and tourism income may consequently lead to rising unemployment rates and economic upheaval. Such phenomena may also have the consequence of increased infrastructure costs in the region, which enlarges the financial burden on local governments. Hence, extreme weather phenomena tend to lower the quality of life for the inhabitants in the area.

As is the case all over the world, the effects of climate change are being felt in Nevşehir. While temperatures are rising, the amount of precipitation is decreasing, the number of snowy days is decreasing, and the number of extreme weather events such as heat waves and floods are increasing [6]. However, if climate change is not successfully combated globally by reducing greenhouse gas emissions, the world's average temperature will continue to rise. When climate modelling studies conducted for Türkiye are examined, it is estimated that temperatures in the Central Anatolia Region of Türkiye will increase by 2.5-3 ºC in 2025 [8, 9]. In 2100, the temperature increase in the region is expected to be 4-6 ºC [9, 10]. Climate modeling for this region also shows that drought periods will be longer and precipitation will decrease [11]. All these effects will increase the climate sensitivity of Nevşehir and inevitably, with the effect of water scarcity, will negatively affect social and economic activities in Nevşehir. One of the human activities that will be affected is tourism, which is the region's most important source of income. The increasing effects of global warming, temperature increase and heat waves (temperatures exceeding 5 °C compared to normal temperatures) will cause a decrease in the number of tourists coming to the region, a decrease in the demand for alternative tourism activities such as horseback tours, ATV tours and balloon tours in the region carried out in open areas, and shifts in the tourism season. On the other hand, important natural and cultural heritage assets such as the famous fairy chimneys and ancient churches of the Cappadocia region will be damaged or destroyed by floods and erosions caused by sudden rainfall. Although this may seem like a doomsday scenario, for example, in 2024, a fairy chimney in the Avanos district of Nevşehir collapsed due to floods and caused material damage to its surroundings [12].

One of the important factors affecting the capacity of Nevşehir province to combat climate change is the rural and urban settlement structure. For example, the use of natural stones, which are frequently encountered in structures built with regional and local architecture, both increases sustainability and creates a structure compatible with climatic conditions (Figure 2) [13]. These structures built using natural materials both save energy and create an architecture compatible with the environment. The diversity of agricultural production in the region is another important factor that supports sustainability and economic development in rural areas. On the other hand, the rapid increase in the urban population, the unplanned urbanization and infrastructure deficiencies triggered by this limit the effective use of the above-mentioned advantages. Especially in recent years, the increasing pressure of tourism has increased the number of boutique hotels and there are a total of 742 hotels in the city, mostly in the Avanos and Ürgüp districts. The rapid construction in the city also makes it difficult to protect natural and cultural heritage assets. As a result, unplanned urbanization causes a decrease in the density of green areas, a decrease in water resources and the degradation of the ecosystem [14]. In addition, the lack of thermal insulation in old buildings constructed before the Energy Performance Regulation in Buildings issued by the Ministry of Environment, Urbanization and Climate Change of the Republic of Türkiye came into force also increases the carbon footprint of the city (Figure 2) [15]. On the other hand, the preference for water-intensive methods in agricultural production in rural areas and the inability to grow agricultural products suitable for the structure of the region both strain water resources and put agricultural sustainability at risk [16]. All these factors listed are elements that will make Nevşehir both resilient and fragile in the process of adaptation to climate change. The production of buildings with high insulation capacity and the continuation of traditional methods in agriculture can increase the city's resistance to climate threats. However, the continuation of unplanned urbanization and excessive resource consumption in Nevşehir will also increase the city's vulnerability to climate threats. For these reasons, it is vital to prioritize climate change combat and adaptation policies and to make strategic plans in this direction in order to protect the natural and cultural heritage in Nevşehir. The preservation of the heritage assets of the Cappadocia Region, such as the famous fairy chimneys, rock-carved churches, underground cities and valleys, the majority of which are located in Nevşehir, is important not only for environmental reasons but also for the sustainability and economic resilience of tourism in the region. For this purpose, a holistic sustainability strategy and policies focusing on the heritage assets of the region should be implemented.

Nevşehir as a Tourism Center in the Cappadocia Region

Nevşehir is located in the center of the Cappadocia region, known for its natural and cultural heritage assets. The fairy chimneys formed by the erosion of volcanic rocks have made the region a center for cultural tourism. Some of the important heritage assets of the region are also included in the UNESCO World Heritage List. The Cappadocia region, known worldwide, is among the important tourism centers in Türkiye with its fairy chimneys, underground cities and rock-carved churches. The city's economy is also largely based on tourism. Unique tourism activities such as cave hotels, hot air balloons, horse and ATV tours have made Nevşehir a center of attraction both nationally and internationally. However, excessive tourism in the region strains the region's resources and threatens its natural ecosystem. Therefore, the implementation of environmentally friendly sustainable development policies is of vital importance both to support the local population and to protect rural life and to prevent migration from rural areas to cities [17]. In this context, the measures to be taken against climate change in Nevşehir should primarily target sustainability.

The Importance of Protecting Natural and Cultural Heritage in Cappadocia

Cappadocia, as it is strategically located, bears the traces of many civilizations throughout history. The region bears traces of the Hittites, Romans, Byzantines, Seljuks and Ottomans. Thus, the cultural heritage of the region is multi-layered. The ancient churches, monasteries and underground cities with frescoes carved into rocks are drawing the religious and cultural-historical richness of this region. These are much more than historical monuments. They are living cultural resources that reflect the past lifestyles, belief systems and artistic understanding of the same Cappadocia region. These heritage assets particularly bring intellectual tourists. However, these heritage assets of Cappadocia are also subjected to climate threats. For instance, floods and erosion destroy cultural heritage assets, making it difficult to pass them on to future generations [18]. Especially fairy chimneys in Cappadocia are eroded by the effects of climate events [8]. People's unconscious actions also cause them to be damaged. Today, the inadequate protection of these heritage assets due to increasing tourism pressure increases the risk of their extinction over time. The protection of these heritage assets and their transfer to future generations is extremely important for the social and economic development of the Cappadocia region. Tourism is the most important income-generating sector of Nevşehir, as well as the most important source of income for the local people, especially in the districts. However, damage and decrease in the heritage assets in the region can also cause a serious loss in the tourism sector. This situation will shrink the local economy and reduce the level of employment in the region. In other words, the protection of the cultural heritage of the region is economically important, but it also serves as a sign of the history, culture and religion of the people as a whole [19].

SUSTAINABILITY AND COMBATING CLIMATE CHANGE IN NEVŞEHIR

The effects of climate change are much more evident in continental regions such as Nevşehir. Therefore, in order to ensure sustainability and adaptation to climate change, comprehensive and innovative solutions need to be implemented in critical areas such as energy, transportation and waste management. In this study, attention is paid to the natural, cultural and economic characteristics of Nevşehir and the focus is on energy, transportation and waste management systems.

Energy: Potential Use of Renewable Energy Resources

There is a great potential for renewable energy resources in the Central Anatolia region due to its geographical location and climate characteristics. In this context, solar energy is a key resource for the region to achieve sustainable energy solutions [23]. Nevşehir, one of the many regions with the highest solar energy potential in Türkiye, has a favorable place in energy production due to the amount of sunlight it receives throughout the year. Utilizing this potential plays an important role in meeting the energy needs of the region and reducing carbon emissions from production practices.

Figure 2. Nevşehir city center and a traditional stone house in Ürgüp, Nevşehir

In Nevşehir, individual and collective projects should be supported to improve the use of solar energy. The application of solar panels to homes and public buildings will reduce the energy dependency in the region. In addition, solar-powered irrigation systems can also help to be an environmentally friendly energy alternative for farmers without increasing costs. Nevşehir is also a place where geothermal energy has potential. Geothermal energy can be used for heating in homes or even greenhouses. Therefore, increasing the development and use of geothermal energy can help reduce the carbon footprint caused by fossil fuel consumption in the region. In order to develop renewable energy projects, these can be supported with financial incentives at local and national levels. Local governments should also raise awareness in the society to support renewable energy. Energy cooperatives can also be established to ensure that local people in the region actively participate in energy production [24]. In addition, it is very important to have energy-efficient technologies and to build green buildings to increase energy efficiency in the city.

Transportation: Bicycle Lanes, Walking Areas and Public Transport Incentives

In order to switch to an environmentally friendly transportation system, both urban and rural transportation solutions should be developed sustainably in Nevşehir. This will contribute to the future of the region. Sustainable transportation will reduce the negative impacts on the environment while also improving public health and quality of life.

The creation of bicycle paths and walking areas in Nevşehir will encourage environmentally friendly transportation modes. Rental bicycles can be an attractive alternative for sightseeing in touristic areas. Such practices will also encourage local residents to use bicycles for short distances. The walking areas to be created will encourage locals to live a healthy lifestyle and will also attract tourists as alternative sightseeing areas in Cappadocia.

In order to reduce the use of private vehicles, it is important to establish public transport systems in Nevşehir and to include electric buses in public transport systems. The implementation of such "green" public transport systems will not only save fuel in the city but also reduce carbon emissions [25]. Increasing the number of such systems will increase the frequency and accessibility of the public transport system and thus guide the society towards this goal. In a tourist region such as Cappadocia, another important application will be the implementation of modern transport solutions such as electric buses or light rail systems that can help tourists visit various areas. Such tourism infrastructure provisions will also increase the environmental sustainability dimension of tourism in the region. As for individual vehicle use, the establishment of more charging stations for electric vehicles in the region can increase the number of environmentally friendly vehicles. The implementation of tax incentives for electric vehicles in the country will also accelerate this transition process.

Waste Management: Local Waste Recycling Projects and Zero Waste Policies

More than 4 million people visited touristic areas in Nevşehir in 2024. The increase in the number of visitors in the region also increases the amount of waste to be generated. Therefore, waste management in touristic areas is important in terms of sustainability. In this context, zero waste policies are also implemented in Nevşehir and materials such as paper, plastic, glass and metal waste are collected separately at the source and recycled. With this circular economy application, natural resources are protected, energy is saved and greenhouse gas emissions are minimized. Public participation in such environmental practices and the expansion of recycling projects will not only maximize the amount of waste collected, but will also make a significant contribution to sustainable development. The implementation of zero waste policies in Nevşehir will contribute to the increase in waste collection, separation and waste recycling facilities and thus increase employment. On the other hand, composting of organic waste from the kitchens and animal farms of households and businesses in the region is important in terms of agricultural practices [26]. The organic fertilizer obtained from this composting process will increase soil fertility in Cappadocia and reduce the use of chemical fertilizers. Thus, an environmentally friendly form of agricultural production can be offered.

GOOD PRACTICE EXAMPLES ON A GLOBAL SCALE

Successful international examples that have fought against climate change will be a very good source for Nevşehir to innovate and develop solutions that can be implemented. The different ways of coping with climatic changes by different regions can be modified for Nevşehir with respect to local needs and conditions. In this respect, it should be very useful to look at the successes from both national and international cases in areas such as energy, transportation and waste management to formulate strategies for application in Nevşehir.

Germany: Renewable Energy Cooperatives in Rural Areas

Germany has drawn attention to investments in renewable energies as part of the fight against climate change [27]. In addition, energy cooperatives established in rural areas of the country have been effective in raising people's social awareness and involving them in renewable energy projects [28]. Energy production from renewable sources such as solar, wind and biomass has become possible through these cooperatives and has been used to meet the energy needs in rural areas. These projects also stimulate the rural economy and facilitate rural development.

Copenhagen, Denmark: Climate Neutral City

Copenhagen is one of the few cities aiming to be carbon neutral by 2025. It is making great strides in this direction [29]. Its most popular infrastructure project includes the promotion of bicycles and electric bus systems in public transport [30]. Wind energy also meets a significant portion of the city's energy needs.

San Francisco, USA: Waste Management and Zero Waste Policies

While many cities fail to implement zero waste policies, San Francisco has made it possible by recycling over 80% of its solid waste and composting all organic waste [31]. Banning single-use plastics and raising awareness about recycling are some of the key policies that have supported this success [32].

Freiburg, Germany: Sustainable Urban Planning

Freiburg stands out among cities that have taken initiatives to become sustainable through innovatively planned urban changes [33]. Citizens have largely engaged in green forms of mobility through the deployment of solar energy systems, restrictions on vehicle traffic, and initiatives to get people to use more bicycles. Energy-efficient buildings also contribute greatly to keeping emissions in the city low [34].

Vienna, Austria: Green Space Management and Urban Agriculture

Vienna is an excellent model city for sustainable practices that promote independent spaces and urban agriculture within the urban fabric [35]. The aforementioned microclimates and biodiversity are enhanced by these areas in Vienna, as comprehensive plans for the expansion and preservation of green areas are underway. Vienna also pursues policies to protect agricultural land within the city and promote local food production [36]. Important structures integrated into the city's planning include bicycle paths, hiking trails and environmentally friendly public transport systems [37].

Melbourne, Australia: Green Infrastructure and Heat Island Mitigation

Melbourne is one of many major cities that have earned the title of “Green City” through heat island reduction programs [38]. The expansion of urban forests has been achieved through tree planting campaigns in the city. These green spaces help reduce perceived temperatures in the city and improve air quality [39]. Rain gardens have also been established throughout the city to naturally filter and reuse rainwater [40]. Melbourne has engaged the public in such green infrastructure projects through awareness campaigns.

Stockholm, Sweden: Climate Neutral and Green Infrastructure

Stockholm is one of the most sustainable cities in Europe. The city has also created wildlife corridors, pioneered in protecting water resources and greening sewer systems [41]. As part of its carbon emission reduction targets, it also invests in energy-efficient buildings and renewable energy projects.

Adaptability of These Examples to Nevşehir

Adapting some of the successful global examples mentioned above to the natural and cultural characteristics of Nevşehir increases the economic and environmental sustainability of the region while also causing local problems to emerge. The solutions in these examples should address the needs of the local people, the existing infrastructure and the local economic framework in their entirety. For example, the socio-economic characteristics of the region should also be taken into account for the renewable energy cooperative model, which has been adopted very successfully in Germany and can be easily adapted to the high solar energy potential in Nevşehir.

Germany's Renewable Energy Cooperatives

Nevşehir has a very suitable climate and solar radiation for the development of solar energy projects [42]. Therefore, the transformation of rural areas in Nevşehir can be achieved by adopting an energy cooperative model from Germany. Germany's renewable energy cooperatives have enabled communities to jointly own and manage renewable energy projects, thus distributing benefits equitably and drawing public attention to clean energy transitions. This model is also suitable for the sustainable development of Nevşehir, where rural development and sustainable energy production are priorities. Solar cooperatives will provide rural communities in Nevşehir with opportunities to produce energy for their own consumption and develop a sense of ownership and mutual sharing of responsibility. This will increase the financial viability of solar energy projects while also improving social cohesion and infrastructure resilience. In addition, the benefits of renewable energy through the cooperative model include job creation, reducing energy costs, and improving the rural population by keeping the income from the sale of excess energy in the local economy [43].

The continuation of state subsidies for renewable energy, which are also available in Türkiye, low-interest loans for cooperative members and feed-in tariff schemes will increase the number of small and large-scale enterprises. In addition, implementation of capacity building programs for stakeholders in cooperative governance, renewable electricity technologies and financial management will be important for the long-term sustainability of these enterprises. Therefore, the German renewable energy cooperative model, transformed to suit the specific context of Nevşehir, has the potential to provide multiple benefits such as energy sustainability, rural development and economic diversification. The abundant solar energy in this region and the participation of the local people in the energy transformation will enable Nevşehir to progress in sustainable rural development and renewable energy production. The intensive energy demand of touristic facilities in the region can be an opportunity for the implementation of renewable energy models.

Copenhagen's Transportation Model

Copenhagen’s extensive network of bike paths and efficient electric public transport have been an exemplary application of sustainable transport infrastructure [44]. This transport system has served both locals and tourists, reducing transport-related environmental pollution to the lowest possible level. By implementing such successful models in Nevşehir, a more sustainable tourism environment that reflects the environmental and economic goals of the region can be achieved. The special geography and unique landscapes of Cappadocia provide an excellent natural reason for an extensive network of bike paths. Bike paths can also be used to connect some important tourist attractions such as the Göreme Open Air Museum, Uçhisar Castle and Love Valley. In this way, tourists can be encouraged to engage in physical activity while reducing their carbon footprint. The creation of rental facilities and bike-friendly paths for both electric and conventional bikes can increase their desirability among visitors. In addition, integrating bike paths into Cappadocia’s tourism infrastructure can reduce vehicle traffic. This means that tourists perceive the region as calmer and more livable.

Electric buses can be used to travel between historical sites and museums in Nevşehir. Such a bus service will provide a clean, efficient and reliable alternative for tourists [45]. Routes connecting Nevşehir city center to important touristic sites in the region can be provided by electric bus service. Renewable energy sources can be used to power electric buses. This will also be an environmentally friendly way of generating electricity. Electric public transport will not only reduce greenhouse gas emissions [46]. In addition, noise pollution from transportation will be significantly reduced. This is also important for the protection of heritage sites in Cappadocia and the sustainability of tourism. On the other hand, digital applications for online ticket sales, real-time route and schedule information, and digital infrastructure should be prepared to increase user convenience and satisfaction. Similarly, sustainable travel models provide significant advantages, even if they are not directly related to the environment. For example, the development of bicycle paths and electric bus services will provide employment opportunities for local people in transportation management, maintenance and rental services. On the other hand, environmentally conscious intellectual tourists, also known as "green tourists", seek destinations that focus on sustainable development [47]. The construction of bicycle paths and electric public transport will make Cappadocia a cluster point for sustainable tourism, attracting intellectual and environmentally conscious visitors. This will increase the international promotion of the region and contribute to a sustainable increase in tourist numbers and income over time. However, the cost of infrastructure, user safety and comfort, and adaptation to existing transport systems are the most challenging issues. Therefore, a comprehensive feasibility study should be conducted before full implementation and pilot programs should be implemented to assess the economic viability and environmental assessment of the initiatives.

San Francisco's Zero Waste Policies

Effective waste management is a vital part of sustainable development for tourist destinations such as Nevşehir and Cappadocia, which host millions of tourists each year. Learning from San Francisco's successful zero waste campaign will be insightful in addressing the challenge of waste accumulation in high-traffic areas. These can then be adapted to the specific socio-economic and environmental context of Nevşehir for sustainability and the preservation of Cappadocia's natural and cultural heritage.

Tourism in Cappadocia generates a high amount of organic waste, especially by reducing activities from hotels, restaurants and the local market. Creating some kind of composting site or facility in this area would provide a sustainable solution to such large organic waste. Composting organic waste reduces the amount of waste sent to landfill and produces a nutrient-rich marketable compost that is ideal for agricultural application in Nevşehir. This approach is very suitable for the heavily agriculturally dependent region, making the soil more fertile and reducing the dependence on chemical fertilizers. Local partnership training programs with local businesses can also be used to separate organic waste at source to improve the quality of the compost produced. Decentralized compost units in rural areas can increase opportunities for communities within community-based waste management and job creation and promote rural development. San Francisco's zero waste strategy is largely based on recycling infrastructure, an important element that could also be implemented in Nevşehir [48]. In addition, easily accessible recycling bins in touristic areas such as Göreme, Uçhisar, Avanos will greatly encourage tourists to separate their waste. Bins should be clearly marked for different wastes, including plastic, metal, glass, paper and others, for easy recycling.

Another important issue is to reduce waste consumption. Advocating and promoting environmentally friendly practices such as water bottles and biodegradable packaging materials, re-bottling, water bottles, eco-friendly straws, etc. can be taken as an example. This can be done by encouraging tourism operators and local businesses to adopt plastic-free practices through tax incentives or certification. Public campaigns targeting tourists and local residents will inform them about the harmful effects of plastic waste and change their behavior. However, popular tourist spots in Cappadocia often lack an effective and modern waste collection infrastructure. Therefore, solid waste makes tourist areas more polluted. It is very important to increase the number and location of waste collection points in these areas. Smart waste bins equipped with sensors should report their filling levels, thus optimizing the collection schedule and emptying the bins before they overflow. Technology will increase efficiency in waste management and also contribute to providing visitors with a better tourism experience by cleaning up areas with heavy traffic. Tourist information is one of the key components of sustainable waste management. Information campaigns, signage and digital platforms can inform visitors about recycling practices and the importance of waste reduction control and preservation of Cappadocia's natural beauty. Some interactive programs, such as guided tours of local recycling and composting facilities, can also engage tourists and foster a greater appreciation of the region's commitment to sustainability. Hotels and tour operators, on the other hand, offer eco-friendly facilities. Zero-waste policies are implemented with a supportive policy framework. Local governments, particularly in the tourism sector, are making effective use of regulations requiring businesses to separate and recycle waste.

Freiburg's Sustainable Urban Planning

The city of Freiburg in Germany is known worldwide for sustainable urban planning, energy efficiency, and efficient public transport systems [34]. Such practices will increase the city's potential to adapt to specific needs and opportunities in Nevşehir. Energy-efficient building design combined with infrastructure development to support electric vehicles (EVs) will provide additional benefits to Nevşehir's urban planning strategies, aiming to learn from international sustainability goals while also addressing many other local issues. Another element of Freiburg's urban planning framework is energy-efficient building design, which aims to achieve energy savings through renewable energy connections [49]. This can be applied to new urban development in Nevşehir. Energy-efficient designs in Nevşehir may include the installation of high thermal performance building materials, insulation systems, and photovoltaic panels to take advantage of solar energy, the most abundant resource in the region. Retrofitting old buildings in Nevşehir with energy-efficient technology will be another improvement. For example, the historic structures of the area can be updated with minimal changes (in many cases, simply by installing energy-efficient systems), while this cultural and architectural heritage can be preserved in a way that maintains as low an environmental footprint as possible.

Freiburg has designed a sustainable transportation model that includes an extensive network of EV charging stations that enable electric mobility for both residents and visitors [50]. Nevşehir could take such an initiative to encourage the transition to electric vehicles and thus contribute to Türkiye's overall goal of reducing carbon emissions and promoting renewable resources. Increasing the presence of EV charging stations in the urban area of Nevşehir and in major tourist areas such as Göreme and Uçhisar would create momentum for the transition to electric mobility for both locals and travelers. The installation could be supported by solar charging systems that take advantage of Nevşehir's high solar energy potential, thus creating a closed clean energy loop for mobility. The charging infrastructure could be improved by including not only fast charging stations for short stops but also standard charging stations in places such as hotels, restaurants and parking lots. Furthermore, electric public transport vehicles such as buses and minibuses could be preferred to minimize noise and atmospheric pollution in the city and its surroundings. This transition could be made possible by establishing partnerships with the private sector specialized in the development and management of EV infrastructure, and the systems could be made an effective and economical alternative in the sector. Such measures not only provide environmental benefits, but also economic benefits. The economic benefits of green building and institutional infrastructure development include jobs in construction, technology, and infrastructure maintenance. More importantly, ending energy consumption without ending non-motorized transportation could help Nevşehir gain international recognition as a sustainable city, thus attracting environmentally conscious tourists and investors.

Vienna's urban agriculture practices

Vienna's innovative approaches to urban agriculture and green spaces can guide Nevşehir in advancing its sustainable strategy. Adopting and adapting these approaches will enable Nevşehir to tackle agricultural production and sustainable urban challenges while improving environmental quality and tourist experience in some of its key locations, such as the fairy chimneys in Cappadocia.

Urban agriculture in Vienna addresses three important issues, namely local food production, community participation and environmental sustainability [51, 52]. Having said this, Nevşehir may also have urban farms that aim to increase productivity and resilience in local agriculture. For example, developments could include rooftop farming, vertical gardens or even small urban greenhouses scattered throughout the city, from rooftop oases to urban gardens, and these greenhouses aim to produce both food security and an environmental footprint reduction strategy for the transportation of this food [53]. Urban agriculture in Nevşehir will depend on the climatic and geographical characteristics of the region. Advanced techniques such as hydroponics and aquaponics can serve to intensively increase productivity in limited urban areas. These systems also use much less water than traditional farming systems, which is appropriate given that Nevşehir has a semiarid climate where water resources are often low. Urban agriculture projects could also include the cultivation of native and drought-resistant plants for broader sustainability and environmental compatibility [54]. Community involvement is what makes or breaks urban agriculture programs. Local small-scale initiatives can motivate people to be more aware of activism and further increase their sense of ownership and community focus. Partnerships with local schools and educational institutions are most directly involved in integrating urban agriculture into their curriculum, thereby modeling sustainable practices and food production.

Green areas are friendly places that have become important within the urban system to improve urban microclimates, reduce the heat island effect and provide aesthetic and ecological benefits [55, 56]. Creating more green areas around touristic areas such as the fairy chimneys in Nevşehir can improve microclimates and provide an overall better experience for tourists visiting the area. Properly maintained greenery has the ability to provide natural shading, reduce ambient temperatures and improve air quality in these crowded hot spots [57]. One solution that urban planners can intervene in is the development of green corridors and buffer zones surrounding major touristic areas such as archaeological sites. These would be areas characterized by the use of local plants and trees that are specifically adapted to the climate and therefore require little maintenance and water use. Buildings and institutions that host tourists can further complement this by having green walls and roof gardens. The expansion of green areas around touristic areas, which further improves the microclimate, would also serve as a visual and experiential upgrade for visitors. This would be walking paths with local vegetation and resting places in the shade. By providing greater comfort to tourists, it can encourage them to stay in the area longer and engage more deeply.

Combining urban agriculture and tourism is a major development opportunity for Nevşehir. These agritourism venues—from tourists harvesting grapes to learning how their ancestors farmed—can add another appeal. They can be among the most effective ways to promote ecotourism while benefiting local farmers and the community. Urban agriculture exhibitions around tourist areas can also demonstrate a growing commitment to making the site relevant to visitors to Nevşehir. For example, demonstration farms that provide eco-parks can serve to showcase new farming technologies while preaching environmental stewardship, thus working toward establishing the region as a premier eco-exotic destination.

A successful urban agriculture and green space strategy in Vienna requires strong policy support and cooperation between different stakeholders [58]. Local governments in Nevşehir could allocate unused or underutilized urban land for agricultural purposes and offer incentives for rooftop or vertical farming. In addition, this could encourage and stimulate widespread participation by providing tax benefits, grants or subsidies to businesses and individuals involved in urban agriculture. Zoning regulations requiring a green buffer could facilitate green space initiatives that limit a tourist area to such new developments. Such private tourism operators could be encouraged to allow green space on their premises. Public awareness campaigns on urban agriculture could be conducted and public support for programs working on green space could be built. Such promotional activities could, for example, communicate the benefits of urban agriculture and green space to the wider community, so that residents and tourists can find meaningful ways to involve themselves in these programs.

In conclusion, combining urban agriculture with improved green spaces in the Nevşehir region has great environmental and economic benefits. Urban agriculture reduces dependence on imported food, reduces carbon footprint and increases biodiversity [59]. Green spaces provide habitats for local wildlife, leading to improvements in air quality and reduction of heat stress [60]. All these initiatives promote a healthier urban environment and the sustainability of the same environment. In particular, they have economic implications in terms of the availability of jobs in urban agriculture and green spaces (such as farming, landscaping and tourism) and improving the quality of the external economy to attract more people to visit.

Melbourne's Afforestation and Rain Garden Projects

Melbourne afforestation schemes and rain garden initiatives provide excellent examples of urban greening and water management that can be adapted to the hot summer climate realities of Nevşehir. When tailored to the specific environmental and socio-economic context of Nevşehir, these measures can dramatically increase greening, promote climate resilience and enhance the development of tourism and public health benefits. Much of the afforestation in Melbourne has focused on improving urban green cover, which tends to reduce the urban heat island effect, improve air quality and offer a range of ecological and aesthetic benefits. A similar initiative could be undertaken for afforestation in Nevşehir, with regard to hot summer climates and semi-arid environments.

Afforestation in the most popular cities and tourist areas such as Göreme, Uçhisar and Avanos may involve planting some local, drought-resistant tree species such as juniper, pine or local fruit trees. The use of these species requires little water and maintenance but can absorb a large amount of shade and coolness. Instead, green belts around towns and buffer zones adjacent to tourist areas such as fairy chimneys will serve the potential of natural cooling, providing more comfort to residents and tourists during the hottest summer months. Afforestation can beautify and increase the aesthetic value of any tourist area such as Nevşehir. Such striking landscapes will attract tourists who want an environmentally speaking experience and those looking for outdoor recreation. Tree-lined paths, shaded rest areas and urban parks associated with local architecture create an attractive blend of culture and nature for tourists and families.

Rain gardens are an important feature in Melbourne's water-sensitive urban design [61]. They capture, filter and store rainwater runoff to minimize flooding and improve water quality in cities [62]. Such systems can be modified for urban and rural areas of Nevşehir, which are in great need of efficient water management due to limited water resources and semiarid climate conditions. In Nevşehir, rain gardens should be constructed at relevant locations throughout urban areas to capture runoff from streets, rooftops and other hard surfaces. These gardens will be filled with native plants that have high water absorption and filtration capabilities and are effective in removing pollutants and sediment from runoff before it enters local water bodies. Rain gardens will also contribute significantly to improving water sustainability in Nevşehir, primarily by replenishing groundwater and reducing surface water losses. They can even be incorporated into public spaces, parks and tourism facilities and serve as aesthetic enhancements and functional water management. For example, they can be placed as educational tools near major tourist attractions to demonstrate Nevşehir's commitment to sustainability and environmental innovation.

Expanding greenery in the form of afforestation and rain gardens has significant public health benefits. More vegetation will reduce the volume of pollutants in the air, alleviate the pain of heat stress, and increase physical and mental well-being. Furthermore, urban green spaces encourage a greater variety of outdoor activities, leading to healthier lifestyles and improved social interaction between residents and tourists. In the contexts mentioned above, the cooling effect from afforestation and rain gardens may be most beneficial for vulnerable populations, such as the elderly and children, who are at greatest risk of exposure to very high temperatures. Rain gardens will improve the quality of water, thereby reducing the incidence of waterborne diseases and creating better health for the entire community [63].

Nevşehir could also open up new economic opportunities as afforestation and rain gardens create jobs in landscaping, horticulture and environmental management to establish and maintain green infrastructure. Furthermore, the increased aesthetic and ecological value added to cities can lead to greater tourist appeal, which in turn can lead to increased revenues for businesses and the hospitality sector. These projects can also be approached from an ecotourism perspective. For example, tours interpreting the ecological and cultural significance of afforestation and rain gardens can be conducted to create unique attractions for responsible eco-travellers. Strong policy support and stakeholder collaboration are needed to successfully translate Melbourne's strategies into the interior. Nevşehir local governments can enact urban planning regulations requiring new developments to have a green infrastructure element and provide financial incentives for afforestation and rain garden projects. Public-private partnerships will mobilise resources and expertise to accelerate implementation. A community must be educated and engaged, so that public awareness campaigns can be led and grassroots education can be embedded in the community's involvement in planting and maintaining urban greenery. Schools and universities can include afforestation and rain garden projects to instill environmental stewardship among the younger generation. Monitoring and evaluation mechanisms should be developed to assess the effectiveness of afforestation and rain garden interventions on improved environmental conditions and community well-being.

Stockholm's Ecological Corridor Model

Stockholm's ecological corridor model offers a holistic approach to preserving existing ecosystems in relation to urban development and sustainable management [64]. The model can be characterized by interconnected green areas and natural habitats, which could be an effective strategy to preserve the original landscapes of fairy chimneys and valleys in Nevşehir. Diversity and alternative energy use would support such a model in close harmony with Nevşehir's sustainable development goals.

The fairy chimneys and valleys in Göreme National Park and Ihlara Valley in Nevşehir are internationally recognized. Unfortunately, however, these heritage assets are under serious threat due to urbanization, climate change and excessive tourism. The adaptation of the Stockholm ecological corridor model serves as a guide for the preservation of these natural environments as elements of an environmentally friendly economy and tourism [65]. As a general outline, ecological corridors can also connect national parks with green areas such as valleys and rural environmental areas, providing the possibility of free wildlife movement and maintaining local plant life and ecological processes such as pollination and seed dispersal [66]. In this, critical habitats will be linked to each other in strategies through the construction of natural paths such as river banks, forest areas and agricultural lands [67]. In addition to preserving biodiversity, such corridors act as natural barriers against urban sprawl while preserving the aesthetic and ecological integrity of Nevşehir's iconic landscapes [68]. For example, by establishing buffer zones with natural vegetation around fairy chimneys, human impact will be reduced and mitigated, thus ensuring the long-term sustainability of these places. Stockholm's ecological corridor plan is more about protecting wildlife and the environment than infusing it with sustainable energy uses towards larger development goals. Similarly, Nevşehir will have to protect these values as well, in order to increase energy efficiency and promote renewable energies within the boundaries of the ecological corridor.

Solar and wind energy projects can be included in the management of the corridors as one of the most abundant sources of solar radiation and wind to be utilized in Nevşehir. For example, small solar systems can provide energy to create ecologically friendly visitor centers, lighting and other infrastructures in the corridors. Similarly, wind turbines that are not installed in sensitive viewing areas will increase the renewable energy capacity without compromising the aesthetic and ecological value. Green architecture such as passive solar design, energy-saving materials and natural ventilation will further increase the energy efficiency of buildings and facilities in the corridors. These developments will consume less energy and use renewable energy sources in line with sustainable development actions for Nevşehir. Such ecological corridors can be possible specific attractions within eco-tourism, one of the fastest growing segments in the travel industry worldwide. Other activities to experience these green corridors can be done by guided tours, walking or cycling, and one can really immerse oneself in the natural and cultural heritage of Nevşehir. Even awareness activities to be carried out in this corridor alone can increase awareness about the biodiversity found in this region of Nevşehir, the importance of sustainability and the need to protect sensitive ecosystems.

Another important aspect of the ecological corridor is local community participation. Involving local residents in planning and management processes means that conservation initiatives are more likely to adapt to local needs and priorities. Community activities that can be organized include reforestation programs and habitat restoration projects that create jobs, increase social cohesion and promote environmental stewardship. Such measures in ecological corridors in Nevşehir require major policy support as well as deep interdisciplinary synchronization of actors across the processes, including local government institutions, local communities, tourism operators and environmental organizations. Environmental corridors provide protection not only for the environment but also for economic sustainability [69]. They increase biodiversity, reduce the effects of climate change and improve ecosystem services such as water filtration and carbon sequestration. Economically, they provide opportunities for ecotourism and renewable energy investments that will generate income for local communities and businesses.

SUGGESTIONS FOR NEVŞEHIR

Sustainable development in Nevşehir and the improvement of its potential in combating climate change require the creation of strategies in critical areas such as energy, tourism, water management, rural development and urban planning. The solutions to be taken under these headings are given below, detailing the cause-effect relationship on a scientific basis.

Energy: Establishment of Solar Energy Farms and Energy-Efficient Building Designs

Nevşehir has great potential for the development of renewable energy sources with its high sunshine duration. The establishment of solar energy farms can meet the energy needs of the region in a sustainable way and at the same time, significantly reduce carbon emissions by reducing fossil fuel use. Solar energy investments will not only provide environmental benefits, but will also create employment for local people and contribute to the economic development of the region [70, 71]. Excess energy production can be transferred to the national energy grid, which can provide additional income to local governments [72]. The use of energy-efficient designs and technologies in buildings provides both environmental and economic benefits by reducing energy consumption and heating-cooling costs. Encouraging green building standards in newly constructed buildings and renovating existing buildings with insulation and energy-efficient systems offer solutions that will optimize energy use. This approach will both reduce energy costs for individual users and reduce the total carbon footprint of the region.

Tourism: Climate-Friendly Tourism Planning and Protection of Natural Areas

The worldwide touristic importance of Cappadocia constitutes the cornerstone of the economic sustainability of the region. However, traditional tourism activities cause environmental damage and threaten natural formations with high energy and water consumption [73]. As part of climate-friendly tourism planning, increasing environmentally friendly transportation options (such as electric buses, bicycle paths) [74] and promoting ecological hotels can reduce the environmental impacts of the tourism sector. At the same time, controlling access to natural areas without limiting the number of tourists will prevent damage caused by overuse in these areas [75]. In addition, the protection of natural areas is critical to securing the long-term tourism potential of Cappadocia and maintaining the ecological balance. Implementation of conservation projects in sensitive areas and regular monitoring of these areas will contribute to the protection of natural heritage such as fairy chimneys. These approaches will ensure the sustainability of tourism revenues in the region and ensure that the natural heritage is passed on to future generations.

Water Management: Sustainable Management of Underground Water Resources

Groundwater resources largely meet the agricultural and drinking water needs of Nevşehir. However, climate change and improper water management practices do not renew these resources and increase the risk of water scarcity. Modern irrigation technologies, together with the promotion of water saving practices, are becoming useful for the sustainable management of groundwater resources. In particular, drip irrigation, one of these irrigation efficiencies, provides positive environmental and economic benefits in terms of reducing water consumption when used in agriculture [76]. Water management strategies should simultaneously provide drinking water to urban people with agricultural production [77]. Scaling up the systems can reduce the demand and pressure on existing water resources in the region [78]. They can provide an effective response to meet the water needs of farmers, especially in rural areas. Any activity carried out in water management will have positive effects on both the economy of Nevşehir and the capacity to adapt to climate change.

Rainwater Harvesting

The decrease in precipitation due to the semiarid climate causes serious effects on Nevşehir's water resources. The increase in climate hazards means disaster for the region and makes it very difficult to manage existing water resources sustainably. Despite all the negativities, rainwater harvesting will be the most effective way to combat drought [79]. Rainwater collection systems can be installed in homes, public buildings and touristic facilities, and the dependence on natural water resources can be reduced [80]. Rainwater collected from roofs can be used for secondary purposes: cleaning, garden watering and even agricultural irrigation [81]. At the same time, the pressure on underground and surface water resources will be reduced without affecting the water cycle. Rainwater harvesting will guarantee access to water in the region during drought periods and will create a buffer against the negative effects of climate change [82].

Reuse of Wastewater

Treatment and reuse of wastewater is a critical issue for Nevşehir in terms of sustainable development goals. Since the growing population, touristic activities and agricultural outputs lead to increased water consumption in the region, development suggests sustainable and efficient use of existing ones [83]. Wastewater treated using advanced treatment technologies can then be used in agricultural irrigation, landscaping and even industrial processes [84, 85]. This will also reduce the consumption of water resources. Recycling of wastewater is important for the environment because it not only saves water resources from depletion but also reduces the carbon footprint in the region. Reuse of wastewater develops a sustainable model for water management, which is an economical and environmentally focused solution in both rural and urban areas of Nevşehir.

Rural Development: Supporting Organic Agriculture and Preventing Rural Population Migration

Agricultural production in Nevşehir is the image of the entire economic architecture of the region. However, these traditional forms of agriculture often result in water wastage and loss of productivity. Adoption of organic agricultural activities will increase environmental sustainability through encouraging reduction in the application of chemical fertilizers and pesticides. More market opportunities for marketing organically produced food products will contribute more to the income of the region and thus energize the local economy. Rural-urban migration can be stopped with the intervention of rural development projects. Training farmers in new farming techniques, credit sensitivity of local cooperatives and a little economic motivation can bring together many people in rural areas with better living conditions. These are some ways to revitalize the rural economy and create less pressure on the population in urban centers.

Urban Planning: Increasing Green Areas and Developing Sustainable Projects with the Participation of Local People

In the modern Nevşehir environment, urban planning should ensure environmental sustainability and improvement of quality of life through a green infrastructure system [86]. Increasing green areas in the city will regulate the microclimate and create public socialization areas. Green corridors will encourage pedestrian traffic and cycling to the city while supporting the ecological balance. Securing social and economic livelihood in the region will include the local people in this process. Community projects can increase public interest in environmental problems and also encourage behavioral adoption of sustainability goals. Implementation of these studies in Nevşehir will make the city a more livable city compared to similar tourism centers. Thus, migration from rural areas to the city can be prevented and even rural life can be encouraged [87 - 89].

CONCLUSION

Nevşehir is among the most unique regions of Türkiye with its nature, cultural heritage and tourism potential. Unfortunately, however, problems such as climate change, increasing environmental pressures and unplanned urbanization are now threatening these values in the region. Therefore, a comprehensive and multi-dimensional approach is required to transform into a sustainable urban and rural settlement model. Similarly, a strategic roadmap should be designed for the region to use its resources efficiently in terms of natural resources and climate change adaptation and to combine economic growth with the surrounding environment. An important opportunity for achieving this goal is Nevşehir's renewable energy capacity. Installing solar energy farms will encourage energy-efficient building designs and support local energy cooperatives. All of this creates a cross-cutting approach to energy supply for the region and reduces carbon emissions. Such energy projects not only benefit the environment but can also support social and economic progress through the creation of local jobs. This will increase the transition to a sustainable agricultural and rural development model that will reduce the migration of people from rural areas and also promote environmental sustainability. The promotion of organic farming, the development of modern irrigation systems and the establishment of water saving policies in agricultural production can improve production in the agricultural sector. In addition, economic incentives for cooperatives and rural development will increase farmers' income levels and improve the quality of life of local people. This is one of the sectors that will provide a solid foundation for Nevşehir's continued economic sustainability, tourism and, of course, climate-friendly conditions for managing such tourism. Dominant policies to protect natural-cultural heritage sites such as Cappadocia, develop environmentally friendly tourism concepts and increase tourists' environmental awareness will keep the region's tourist interest alive and minimize environmental impacts. Another promising measure is to advocate innovative solutions in the region's tourism infrastructure, as well as electric and state-of-the-art vehicles. In this context, infrastructure for green projects should be prioritized in urban development planning in Nevşehir. Thus, more green areas will be included to provide a more harmonious microclimate and a healthier environment for residents. Additionally, projects developed with public participation will facilitate the adoption of sustainability goals and increase social cohesion.

Within the scope of combating climate change, Nevşehir’s adaptation to a sustainable urban and rural settlement model will be the result of a series of interconnected measures to be taken in related areas such as energy, agriculture, tourism, water management and even urban planning. Natural and cultural riches make the region privileged for sustainable development. Utilizing these advantages will put Nevşehir in a position to create and implement its own model in combating climate change and will also make it an exemplary city. However, the protection of the region’s natural and cultural heritage should be prioritized in sustainable development issues. Cultural heritage areas such as fairy chimneys, rock-carved churches and underground cities should be included in Cappadocia’s long-term management plans, managed within the scope of sustainable tourism principles by taking climate change into account and visitor loads should also be reduced. Similarly, the survival of traditional knowledge and cultural values in Nevşehir and the preservation of its regional identity are of great importance. For example, the protection of natural areas will not only serve the local ecology but will also increase Nevşehir's general attractiveness in tourism and increase its economic sustainability. For these reasons, the protection of natural and cultural heritage should be taken as an inevitable prerequisite for sustainable development in Nevşehir. Proper use of these advantages has a great potential to develop an exemplary model for Nevşehir in combating climate change and to make it an exemplary region in the country and even internationally.

DATA AVAILABILITY STATEMENT

The authors confirm that the data that supports the findings of this study are available within the article. Raw data that support the finding of this study are available from the corresponding author, upon reasonable request.

CONFLICT OF INTEREST

The author declares that there is no conflict of interest with any individual, institution, or organization in the preparation, evaluation, or publication of this study.

USE OF AI FOR WRITING ASSISTANCE

Not declared.

ETHICS

There are no ethical issues with the publication of this manuscript.

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